SOFTWARE REQUIREMENTS SPECIFICATION

Bicycle, Meal and Mushroom Management System

Document	Software Requirements Specification (SRS)
Project title	Bicycle, Meal and Mushroom Management System
Technology	C# / Windows Forms / Visual Studio
Version	1.1 – Updated to include Mushroom service
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1. Introduction

This document defines the functional requirements for a Windows desktop application that allows an authenticated user to access bicycle, meal, and mushroom services, calculate charges, and view a transaction summary. It follows the classroom whiteboard workflow and records assumptions where exact business details were not supplied.

2. Purpose and Scope

The purpose is to provide a simple system for selecting one of three services—Bicycle, Meal, or Mushroom—and calculating the applicable total. The initial version includes login, service selection, item selection, quantity or rental duration, total calculation, summary messages, navigation, and logout. It does not store transactions or process online payments.

3. Users and Operating Environment

The intended user is an operator using a Windows computer. The application is developed in C# using Windows Forms in Visual Studio and targets .NET 8.

4. Functional Requirements

ID	Functional requirement
FR-01	The system shall display a login screen when the application starts.
FR-02	The system shall validate the username and password before granting access. Invalid credentials shall show an error and allow ano
FR-03	After successful login, the system shall display a service menu containing Bicycle, Meal, and Mushroom options.
FR-04	When Bicycle is selected, the system shall display a bicycle service screen.
FR-05	The system shall allow the user to select a bicycle type and enter rental duration in hours.
FR-06	The system shall calculate bicycle cost as the selected hourly rate multiplied by rental hours.
FR-07	When Meal is selected, the system shall display a meal service screen.
FR-08	The system shall allow the user to select a meal and enter the quantity required.
FR-09	The system shall calculate meal cost as the selected meal price multiplied by quantity.
FR-10	When Mushroom is selected, the system shall display a mushroom service screen.
FR-11	The system shall allow the user to select a mushroom product/type and enter the quantity required.
FR-12	The system shall calculate mushroom cost as the selected product price multiplied by quantity.

ID	Functional requirement
FR-13	The system shall display a summary message containing the selected service/item, quantity or duration, and calculated total.
FR-14	The system shall allow the user to return from a service screen to the service menu.
FR-15	The system shall allow the user to log out and end the current session.

5. Business Rules and Assumptions

• Bicycle total = hourly rental rate × rental hours.
• Meal total = meal unit price × quantity.
• Mushroom total = mushroom product unit price × quantity.
• Totals are displayed in Rwandan francs (RWF).
• Sample bicycle rates: Standard 1,000 RWF/hour; Mountain 1,500 RWF/hour; Electric 2,500 RWF/hour.
• Sample meal prices: Rice and beans 2,500 RWF; Chicken and chips 3,500 RWF; Beef with rice 5,000 RWF.
• Sample mushroom prices: Oyster mushrooms (pack) 3,000 RWF; Button mushrooms (pack) 5,000 RWF; Shiitake mushrooms (pack) 8,000 RWF.

The whiteboard does not specify exact product varieties, prices, or detailed rules. All sample choices and prices are placeholders and must be confirmed with the lecturer.

6. Non-Functional Requirements

• Usability: The interface shall use clear labels and simple navigation.
• Performance: Totals shall be calculated and displayed immediately.
• Reliability: Invalid login details shall not grant access; numeric inputs shall be restricted to valid ranges.
• Compatibility: The application shall run on a supported Windows computer with the .NET 8 runtime.
• Maintainability: Login, menu, and service screens shall be separated into individual source-code classes.

7. Main Application Flow

Start → Login → Choose Service → Bicycle OR Meal OR Mushroom → Select item and quantity/duration → Calculate Total → Display Summary → Back to Choose Service or Logout.

8. Acceptance Criteria

The system is accepted when a valid user can log in, see all three service options, calculate totals for bicycle, meal, and mushroom selections, view a summary for each, navigate back, and log out. Incorrect credentials must not open the service menu.

9. Limitations and Future Enhancements

This version uses demonstration login credentials and hard-coded sample prices. It does not save user accounts or transactions. Future improvements may include database storage, administrator management of products and prices, transaction history, printable receipts, and stronger authentication.

10. Conclusion

This SRS provides the initial requirements baseline for the Bicycle, Meal and Mushroom Management System. It should be reviewed and adjusted to match any additional instructions provided by the lecturer.